

OptiCrop:Smart Agricultural Production Optimization Engine

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dataset stored locally	Difficult to manage large datasets	High
2	Flask running locally	Limited user scalability	Medium
3	Supports only crop recommendation	Limited agricultural assistance	Medium
4	No user accounts	Limited personalization	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports only local users.	Deploy on AWS, Azure, or Render to support thousands of users.
2	Data Storage	Dataset stored in CSV files.	Use MySQL or MongoDB for large-scale agricultural data management.
3	Performance	Single local ML model.	Use model optimization, caching, and cloud computing resources.
4	Security	Basic input validation.	Add user authentication, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and public access	1–2 Months	Increased accessibility and scalability

Phase 2	Real-time weather API integration	2–3 Months	Improved prediction accuracy
Phase 3	Fertilizer and crop disease recommendation system	3–4 Months	Complete agricultural support platform
Phase 4	Mobile application and multilingual support	4–6 Months	Wider adoption among farmers